

Exercise sheet 5: TranSMART

Before starting the exercise: Install TranSMART with Docker, see GitHub.

TranSMART is available at <http://localhost:80>

user: admin, password: admin

1. Overview of the study

- a. Open the tree structure under "Public Studies" > "MIMIC" on the left under "Navigate Terms" and add the following variables to subset 1:

Age

The subset does not need to be saved.

- b. Go to the "**Summary Statistics**" tab. What is the mean and median age of the patients?
- c. What percentage of patients are male or female?

2. Comparison of subsets

- a. Go back to the "Comparison" tab and delete the subset from task 1 by clicking on "Clear Panel."
- b. Now add new variables to the subsets. You can find the variables under "Public Studies" → "MIMIC" → "Clinical Data" → "Diagnosis."

Subset 1: Variable "E78.5" under "Diagnosis" → "Hyperlipidemia"

Subset 2: Variable "I48.0" under "Diagnosis" → "Atrial fibrillation"

- c. Now go to the "Summary Statistics" tab:
How many patients are in each subset and how many patients have both diagnoses?

What is the mean age of patients suffering from hyperlipidemia or atrial fibrillation?

3. Box plot

Perform the task with the **two subsets** "Hyperlipidemia" and "Atrial fibrillation" from the previous task 2.

- a. Go to the "SmartR" tab and select "Boxplot Workflow" from the drop-down menu. Drag the variable "**hematocrit**" under "Clinical Data" → "Lab Values" into the left field "Numerical Variables." Click on "Fetch Data."
- b. Once the data has been loaded, switch to "Run Analysis" and create boxplots with "**Raw Values.**"

- c. What is the mean hematocrit value for patients with hyperlipidemia or atrial fibrillation? For which diagnosis is the mean value higher?

4. Correlation analysis

Perform the task with the subset "Atrial fibrillation." The subset is changed in the "Comparison" tab.

Then go to the "**SmartR**" tab and select "**Correlation Workflow**."

- a. Select the numerical variables "**age**" and "**weight**". Click on "Fetch Data" and then go to the "Run Analysis" tab.
- b. Create a correlation analysis using "Raw Values" and the **Pearson** method.
- c. What is the correlation coefficient and what does it say about the age and weight of patients diagnosed with atrial fibrillation?

5. Survival time analysis

- a. First, perform this exercise with the "Hyperlipidemia" subset. The subset can be changed in the "Comparison" tab.
- b. Go to the "**Advanced Workflow**" tab and select "Analysis" → "**Survival Analysis**" from the drop-down menu.

Create a Kaplan-Meier curve with the following variables:

Time: **age**

Category: -

Censoring variable: **alive**

- c. Now change the subset to "**Atrial fibrillation**." The subset is changed in the "Comparison" tab.
Now perform a survival analysis again as in task 5.b.
- d. Now compare the two Kaplan-Meier curves for patients diagnosed with hyperlipidemia and atrial fibrillation.
How many patients are still alive at age 80?